THE TRAJECTORY LEXICON

Closed Core 1.2

A fail-closed language for consequential discrimination, trajectory mapping, bounded intervention, agency, safety, responsibility, and governed execution.

Core discipline: never erase a distinction consequence supports; never manufacture one it does not; never spend the capacity to detect error, correct, recover, or reconstruct without explicit warrant.

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Developed by Caleb Meadows with Microsoft 365 Copilot

Status: operationally closed core. Issues not honestly resolvable are explicitly classified Fundamental-Open or External-Normative and fail closed for dependent execution.

0. Status and closure doctrine

The Trajectory Lexicon regulates inference and intervention from stimulus through consequence. Version 1.2 closes the former open-problem register at the level of operational grammar. Every listed issue now has either a deterministic procedure, a domain-parameterized procedure, an empirical resolution path, an external normative boundary, or an explicit fundamental-open classification.

Epistemic openness is allowed. Silent execution authority is not.

EpistemicOpen(U) may coexist with OperationalClosed(I)

UnknownFundamental does not imply PermissionToAct

Closure class	Meaning	Action posture
CLOSED	Rule or bounded comparison procedure specified.	Apply rule; audit deviations.
PARAMETRIC	Rule specified; thresholds supplied by domain.	Require declared parameters and provenance.
EMPIRICAL	Resolution depends on observation or experiment.	Keep inquiry open; act only within present warrant.
FUNDAMENTAL-OPEN	No non-arbitrary internal discriminator presently available.	Preserve unknown; block promotions depending on closure.
EXTERNAL-NORMATIVE	Facts alone do not select the legitimate terminal value/authority.	Require legitimate external authorization.

1. Fundamental floor and non-goals

E0: For every x that is, x is whatever x is.

- Existence does not depend on observation or naming.
- Representation(x) does not imply ExistenceAsRepresented(x).
- FailureToRepresent(x) does not imply Nonexistence(x).
- A useful local name does not establish a fundamental kind.
- Operational closure does not imply epistemic completion.

The framework does not claim exhaustive ontology, exhaustive causation, complete future enumeration, universal terminal values, perfect evidence authenticity, or perfect open-world safety proof.

2. Primitive structure

TL = <E,S,G,R,O,Y,M,H,I,A,U,Omega,Sigma,Gamma,Auth,Prov>

Symbol	Domain	Role
E	Existence	Whatever exists; not assumed exhaustively enumerable.
S	State	Conditions distinguished for inquiry.
G	Trajectory	Transitions, branch points, path dependencies.
R	Regime	Scale, resolution, access, operations, time, intention, conditions, purpose, horizon.
O	Observation	Registered interaction/measurement.
Y	Yield	Registered output after interaction/intervention.
M	Representation	Model, description, mathematics, narrative, category.
H	Hypothesis	Representation proposed to explain/predict.
I	Intervention	Deliberate alteration of accessible relations.
A	Agent	System capable of relevant discrimination, selection, intervention, correction.
U	Unresolved	Typed insufficient classification.
Omega	Option space	Accessible consequential trajectories for an agent.
Sigma	Scaffolding	Dependencies, redundancy, reconstruction, substitution, transmission, correlation.
Gamma	Warrant	Evidence licensing promotion/action.
Auth	Authorization	Legitimate permission state for action.
Prov	Provenance	Source/instrument chain supporting evidence integrity.

3. Speech-act and epistemic grammar

Stim -> Q -> Cand -> Hyp -> Assert -> Dir

Type	Meaning	No automatic promotion
Stim	Interesting input.	Question/belief/cause.
Q	Distinction admitted for inquiry.	Confidence.
Cand	Candidate representation.	Preference/support.
Hyp	Representation proposed to explain/predict.	Causal establishment.
Assert	Claim under warrant and regime.	Universal truth.
Dir	Action request under goal/warrant.	Execution authority.

Q(p) does not imply Belief(p); Attention(p) does not imply Confidence(p)

3.1 Epistemic types

Type	Form	Meaning
Observed	Obs(p|rho)	Registered observation.
Yield	Yield(y|I,rho)	Registered output.
Representation	Rep(m)	Constructed model/description.
Prediction	Pred(y|h,I,rho)	Expected yield.
Trajectory	Pred(G|h,I[0:h],rho,Gamma)	Expected path and branches.
Causal	Causal(x,y|rho,Gamma)	Supported causal relation.
Counterfactual	CF(y|do(x),rho,Gamma)	Modeled unrealized intervention.
Operational	Op(a|goal,Gamma)	Action licensed for stated goal.
Unresolved	Unres_k(p)	Typed insufficient classification.

3.2 Promotion engine

Gamma |- p:tau1 does not imply Gamma |- p:tau2

if PromotionRule(tau1,tau2,Gamma,rho) passes: promote; else retain tau1

Failure to promote is not falsification. Scope, regime, provenance, and warrant stay attached to every promoted claim.

4. Consequential discrimination

D_rho(x,y) in {1,0,?}

Value	Licensed	Not licensed
1	Accessible consequential distinction established.	Universal/fundamental difference.
0	No admitted operation distinguished them.	Identity.
?	Discrimination not established.	False/impossible/equiprobable.

D_rho(x,y)=1 => x !=_rho y; D_rho(x,y)=0 => x ~_rho y

4.1 Regime validity closed

R0=<rho0,purpose,horizon,admittedOps,stopRules,revisionRules>

RegimeDelta=<old,new,independentReason,newVulnerability,authorization>

1. Pre-bind the initial regime and admissible search family.
2. Any post-result change creates a RegimeDelta rather than rewriting the original result.
3. A valid change needs an independent reason and either a new vulnerable prediction or a demonstrated measurement defect.
4. A change that only protects a favored model is immunizing and cannot promote it.

Status: CLOSED/PARAMETRIC.

5. Causal promotion closed

Gamma_c=<T,I,R,P,C,A,B,M>

Term	Warrant dimension
T	Temporal ordering.
I	Intervention sensitivity.
R	Replication.
P	Parameter response.
C	Discriminating controls.
A	Alternative constraint/elimination.
B	Boundary prediction.
M	Mechanism/mediation compatibility.

Level	Licensed language
C0	Association or sequence only.

Level	Licensed language
C1	Causal candidate: at least one causal discriminator survives.
C2	Supported causal contributor: multiple independent dimensions survive; major known alternatives constrained.
C3	Robust causal relation within jurisdiction: replicated intervention-sensitive support plus boundary success.
C4	Structurally forced relative to declared model assumptions, not exhaustive ontology.

CausalLevel never implies ExhaustiveCause

Status: CLOSED at the type level; actual causal evidence remains EMPIRICAL.

6. Trajectory grammar and envelopes

G_i=<S_0,I_0,Y_1,S_1,B_1,...,S_h>

D^G_rho(G_i,G_j|h) in {1,0,?}

Immediate equivalence does not license future equivalence. Every action is evaluated for immediate state change, trajectory shift, and transformation of the future decision field.

Gset(a)={expected,favorable,adverse,catastrophic,adversarial,model-failure,recovery,U}

The U branch is mandatory unless exhaustiveness is independently proven in a bounded formal domain.

6.1 Probability discipline closed

Probability type	Rule
P_exact	Use calibrated distribution only with validated statistical warrant.
P_interval	Keep the interval; do not substitute midpoint by default.
P_order	Use ordinal ordering only where supported.
P_set	Evaluate robustly across admissible models/distributions.
P_unknown	Do not fabricate a number or equal weighting.

P_unknown + high irreversible consequence => increase safety state, not probability=0

Status: CLOSED procedurally; estimated probabilities remain EMPIRICAL.

7. Unresolved remainder closed

Type	Meaning	Required response
U_obs	Observation missing/unreliable.	Improve access/measurement.
U_model	No adequate current representation.	Generate candidates.
U_ident	Candidate pair has D_rho=? under the current regime.	Invoke ResidualRefine; seek an admissible discriminator while preserving unresolved status.
U_scale	Relevant scale unresolved.	Traverse admissible regimes.
U_prob	Likelihood unresolved.	Use deep-uncertainty procedure.
U_value	Legitimate values/purposes conflict.	External normative decision.
U_auth	Authority/consent unresolved.	Do not execute dependent action.
U_integrity	Evidence/instrument provenance unresolved.	Fail closed on dependent claims/actions.
U_fund	No non-arbitrary framework discriminator.	Mark FUNDAMENTAL-OPEN; prohibit dependent promotion.

8. Scaffolding, resilience, and recovery

Sigma(G)=<Dep,Red,Rec,Sub,Corr,Trans>

- Dep: critical dependencies.
- Red: genuinely independent redundancy.
- Rec: reconstruction capacity.
- Sub: substitutes.
- Corr: correlated failure exposure.
- Trans: executable knowledge transmission.

Duplication != IndependentRedundancy

8.1 Recovery adequacy closed

RecPath=<trigger,authority,resources,dependencies,data,skills,time,test,rollback,reconstruction>

1. Verify resources exist.
2. Verify recovery does not depend solely on the failed component.
3. Exercise by simulation, drill, or bounded test where admissible.
4. Record restoration time and unrecovered functions.
5. Treat untested high-consequence recovery as degraded, not demonstrated.

Status: CLOSED procedurally; performance remains EMPIRICAL.

9. Agency, identity, sovereignty, responsibility

K_A={distinctions A can reliably make}

Omega_A(t)={accessible consequential trajectories}

Agency_A=Capacity(IntentionalSelection|K_A,Omega_A,rho)

Sov_A=Consistency(Principles_A,Intention_A,Choice_A,Action_A)

Sovereignty is self-consistent authorship of available choice, not control over existence or guaranteed outcomes.

9.1 Identity continuity closed operationally

Id_A(t1,t2|purpose)=<causal continuity,provenance continuity,commitments,control continuity,accountability continuity>

Same-agent treatment is purpose-indexed. Metaphysical personal identity remains FUNDAMENTAL-OPEN and is not needed for operational use.

9.2 Responsibility ordering closed

Resp_A=<K,Agency,Intention,Alternatives,Foreseeability,Constraints,Contribution,Control,Repair>

Responsibility is partially ordered, not universally scalar. New knowledge, ignored triggers, continued control, concealment, and repair opportunities update the vector. Legal or moral blame rules remain EXTERNAL-NORMATIVE.

10. Partial ordering and conflicting purposes

a >=_rho b iff a is no worse on every authorized decision dimension and better on at least one

If neither dominates, incomparability is retained. The engine does not invent a scalar conversion.

HardConstraintFail(a) => Adm(a)=Fail

Hard constraints must have legitimate provenance. A non-dominated conflict among legitimate terminal values is escalated to the authorized human/policy process.

Status: CLOSED structurally; terminal-value selection is EXTERNAL-NORMATIVE.

11. Consent and authority

Consent=<capacity,information,specificity,voluntariness,revocability,scope,duration,verification>

Consent is licensed only for the dimensions and horizon actually established. When direct consent is impossible, proxy authority must have explicit provenance, and the least-agency-reducing intervention consistent with the authorized duty is preferred.

AuthState in {Unrequested,Requested,Granted,Revoked,Expired}

Legitimacy of proxy, emergency, fiduciary, legal, collective, or guardianship authority is EXTERNAL-NORMATIVE, not manufactured by the inference engine.

12. Capture detection closed

Cap=<Exit,Substitute,Audit,Contest,Reconstruct,AuthorityConcentration,DependencyAsymmetry>

Capture risk rises when meaningful exit, substitution, audit, contestability, and reconstructive capacity decline while authority concentration and dependency asymmetry rise. Size or concentration alone does not prove capture.

> T_capture if multiple independent capability dimensions cross declared bounds or a hard sovereignty boundary fails

Status: CLOSED/PARAMETRIC.

13. Safety states, triggers, and admissibility

State	Meaning	Bound response
GREEN	Inside authorized envelope.	Operate and monitor.
YELLOW	Uncertainty/dependency/discrepancy rising.	Slow; increase discrimination.
ORANGE	Large consequence, difficult reversibility, concentration/correlation.	Constrain; sandbox; add redundancy; ready recovery.
RED	Warrant no longer licenses continuation.	Suspend; contain; reassess.
BLACK	Active failure threatens essential function, agents, information, or reconstruction.	Preserve; contain; restore correction; reconstruct.

Trigger	Condition
T_irr	Irreversibility exceeds authorized threshold.
T_cat	Credible catastrophic branch enters envelope.
T_unk	Unresolved exposure becomes decision-dominant.
T_lock	Future correction/option space collapses.
T_dep	Critical dependency concentrates.
T_corr	Failure paths become correlated.
T_agent	Another agent's meaningful agency materially changes.
T_model	Observed yield exits trajectory envelope.
T_capture	Capability/authority becomes dangerously asymmetric.
T_drift	Objective changes without authorization.
T_resp	New knowledge changes foreseeability/responsibility.

Trigger	Condition
T_regime	Regime changes post hoc to protect model.
T_integrity	Evidence/audit provenance loses trust.
T_recovery	Recovery path becomes non-executable.

13.1 Admissibility

Adm(I|goal,rho,Gamma,Auth,Prov,G_history)=Pass

Admissibility includes cumulative trajectory history, affected agents, consent/authority, tail branches, unresolved exposure, reversibility, recovery, dependency concentration, provenance, correction capacity, and less-consequential alternatives.

HighConsequence(I) => DemonstratedOrQualifiedRecoveryPath(I)

14. Unknown unknowns and model-boundary probing

No live candidate set is presumed exhaustive. Inquiry reserves capacity for observations not optimized solely to distinguish current models.

BoundaryProbeValue=NovelDiscrimination/(Cost+Risk+Irreversibility)

- Vary regimes and supposedly irrelevant variables.
- Inspect structured residuals.
- Seek systematic prediction error.
- Treat repeated auxiliary rescue without new vulnerability as YELLOW epistemic state.

Status: CLOSED as search discipline; exhaustive unknown-unknown elimination remains FUNDAMENTAL-OPEN.

14.1 Residual-refine operator

U_ident(H_i,H_j|rho) iff D_rho(Pred(H_i),Pred(H_j))=?

ResidualRefine(H_i,H_j,rho)=argmax_(rho_prime,I in Adm)
RelevantReliableDiscrimination(Pred(H_i,I,rho_prime),Pred(H_j,I,rho_prime))/
(Cost+Risk+Irreversibility)

ResidualRefine does not assume the candidates are exhaustive and does not promote either candidate. It searches admissible regime changes or interventions for a consequence that can turn unresolved identification into a local distinction or local indistinguishability. If no admissible discriminator is found, U_ident is retained.

15. Multi-agent adversaries and trust separation

ProposalAuthority != EvidenceAuthority != AuthorizationAuthority != ExecutionAuthority != AuditAuthority

When deception is credible, provenance and independent measurement become part of Gamma. Self-reported evidence cannot satisfy a hard gate whose failure benefits the reporter unless separately attested.

16. Execution-boundary independence

The Lexicon terminates in typed operational outputs and authorization requirements. It deliberately does not specify a particular enforcement engine, authentication mechanism, ledger, execution governor, or deployment substrate. Those are separate implementation layers whose guarantees must be evaluated independently.

LexiconOutput=<decisionRecord,admissibility,safetyState,authorizationRequirement,recovery,revision>

Any external execution system may consume a Lexicon output only through an explicit interface whose trust assumptions and failure modes remain outside the Lexicon core. The Lexicon cannot treat successful execution as proof that its reasoning was correct.

16.1 Cumulative trajectory operator

CumAdm(I_t)=Adm(I_t|G_(0:t),Sigma_(0:t),Omega_(0:t),Gamma_t,Auth_t,Prov_t)

CumAdm is the single canonical cumulative-action rule. Any section that evaluates sequential interventions references this operator rather than restating the formula. Individually admissible actions can therefore compose into an inadmissible trajectory.

17. Formal implementation contract

Machine	States
SpeechAct	Stim,Q,Cand,Hyp,Assert,Dir
Epistemic	Obs,Yield,Rep,Pred,Trajectory,Causal,CF,Op,U
Safety	GREEN,YELLOW,ORANGE,RED,BLACK
Authorization	Unrequested,Requested,Granted,Revoked,Expired
Intervention	Proposed,Review,Cooling,Authorized,Executing,Sealed,Halted,Recovery
Trajectory	Mapped,Monitoring,Deviated,Revised,ClosedForGoal

17.1 Proof obligations

- No epistemic promotion without typed rule and warrant.
- No consequential execution without admissibility and valid authorization.
- RED cannot execute absent an explicit externally authorized override record and responsibility transfer.
- BLACK permits preservation, containment, restoration, or separately authorized emergency action only.
- Every executed action receives a consequence record; durable anchoring is required where the deployment provides it.
- Every model update preserves provenance of the superseded state.
- Trajectory maps cannot be marked exhaustive without an independent bounded-domain proof.
- Unknown probability cannot silently become a number.

Status: CLOSED as a specification. Machine-checked proofs remain an engineering deliverable, not an unnamed framework gap.

18. Fundamental-open register

Fundamental-open	Why	Operational treatment
Exhaustive ontology	A representation cannot establish an inventory of all existence in general.	No dependent universal claim.
Ultimate cause	Supported cause does not establish exhaustive causal possession.	Use bounded C-levels.
Complete future enumeration	Open systems may contain unrepresented states/agents/innovations.	Mandatory U branch; preserve correction/recovery.
Universal terminal value ordering	Descriptive facts do not derive every legitimate end.	External normative authority resolves hard conflicts.
Metaphysical personal identity	Operational continuity does not settle metaphysics.	Use purpose-indexed identity only.
Perfect evidence authenticity	Verification rests on physical/institutional trust roots.	Expose roots; diversify/attest proportional to consequence.
Perfect open-world safety proof	Proofs are relative to model and specification.	Runtime monitoring, boundary probes, recovery.
Unknown-unknown exhaustion	General search cannot prove nothing lies outside candidate space.	Maintain U_fund and boundary probes.
Legitimacy from facts alone	Authority is not generated solely by descriptive inference.	Separate normative authorization from warrant.

Fundamental-open never means fail-open execution. It means inquiry stays open while actions depending on unavailable warrant fail closed.

19. Closed-core master invariants

1. Existence is not representation-dependent.
2. Speech acts are typed before evidence evaluation.
3. Claims retain type, regime, warrant, scope, and provenance.
4. No promotion without explicit rule.
5. Failure to promote preserves lower knowledge.
6. Candidate sets remain non-exhaustive unless independently proven exhaustive.
7. Every trajectory envelope retains unresolved exposure.

8. Probability precision cannot exceed warrant.

9. Catastrophic exposure is not erased by unknown probability.

10. Hard constraints precede soft optimization when legitimately declared.

11. Normative incomparability is escalated rather than silently scalarized.

12. High-consequence intervention requires proportionate recovery architecture.

13. Safety triggers bind responses.

14. Regime changes preserve earlier results and require independent justification.

15. Cross-action cumulative consequence is evaluated before execution.

16. Credible deception makes provenance part of warrant.

17. Proposal, evidence, authorization, execution, and audit roles are separable.

18. Operational closure never implies epistemic completion.

19. Knowledge/agency changes update responsibility.

20. Unresolved fundamentals cannot become hidden permissions.

20. Closure matrix

Former open issue	Resolution	Status
Regime validity	R0 precommitment + RegimeDelta + anti-immunization.	CLOSED/PARAMETRIC
Causal promotion	Gamma_c + C0-C4 bounded causal language.	CLOSED/EMPIRICAL
Trajectory probability	Typed probability + deep-uncertainty treatment.	CLOSED/EMPIRICAL
Partial ordering	Dominance + hard constraints + normative escalation.	CLOSED/EXTERNAL-NORMATIVE
Unresolved partition	Nine U classes.	CLOSED
Identity continuity	Purpose-indexed continuity tuple.	CLOSED/FUNDAMENTAL-OPEN metaphysics
Responsibility ordering	Partial responsibility vector and updates.	CLOSED/EXTERNAL-NORMATIVE blame
Conflicting purposes	Retain non-domination; authorized terminal choice.	CLOSED/EXTERNAL-NORMATIVE
Consent/representation	Consent tuple + proxy provenance.	CLOSED/EXTERNAL-NORMATIVE authority
Capture detection	Capability-loss vector and trigger.	CLOSED/PARAMETRIC
Recovery adequacy	Executable recovery tuple + testing.	CLOSED/EMPIRICAL

Former open issue	Resolution	Status
Unknown unknowns	Boundary probes + structured residuals.	CLOSED/FUNDAMENTAL-OPEN exhaustion
Multi-agent adversaries	Trust separation + provenance + cumulative trajectory.	CLOSED/EMPIRICAL authenticity
Formal implementation	State machines + transition obligations.	CLOSED as specification
Lexicon self-typing	Closure classes + fundamental-open register.	CLOSED

21. Full operating protocol

1. Register stimulus without promotion.

2. State target consequential distinction.

3. Type speech act.

4. Define observable yield.

5. Pre-bind regime, purpose, horizon, admitted operations, and revision rules.

6. Enumerate candidates plus typed unresolved remainder.

7. Attach evidence provenance.

8. Generate divergent predictions and trajectory envelopes.

9. Map scaffolding, affected agents, option-space change, cumulative history, and recovery.

10. Select strongest admissible discriminator.

11. Pre-bind result meanings, safety triggers, and responses.

12. Evaluate hard constraints, authority, and consent.

13. Pass admissibility.

14. Execute least consequential sufficient intervention through the governed execution path.

15. Register yield without causal overpromotion.

16. Compare observed and predicted trajectory.

17. Evaluate triggers and recovery executability.

18. Continue, constrain, suspend, contain, or recover as bound.

19. Revise models, causal level, probability type, trajectory map, responsibility, and option space.

20. Record boundary, provenance, and unresolved remainder.

21. Close operationally only for stated goal and expiry.

22. Repeat only while expected discrimination exceeds justified exposure.

22. Self-test suite

Test	Pass condition
Question integrity	Question never becomes belief automatically.
Type safety	No unauthorized epistemic promotion.
Regime drift	Post-result changes create RegimeDelta.
Causal restraint	C-level does not become ontology.
Probability restraint	Unknown is not numericized.
Tail visibility	Catastrophic branch stays represented.
Redundancy audit	Shared dependencies exposed.
Cross-action composition	Sequence assessed cumulatively.
Capture blindness	Loss of exit/audit/reconstruction triggers review.
Recovery fiction	Untested recovery not treated as demonstrated.
Deception	Self-report cannot satisfy interested hard gate unaudited.
Normative smuggling	Facts do not invent terminal values.
Operational stopping	Action can close while knowledge remains open.
Self-application	New contradiction reopens relevant section.
Layer separation	Lexicon reasoning does not inherit guarantees from an external execution system.

23. Developmental grounding and status

This closed core preserves the original Trajectory Lexicon commitments: consequence-indexed discrimination; speech-act typing; fail-closed promotion; trajectory mapping; scaffolding; option-space analysis; safety states; recovery; responsibility; and self-application. It also preserves the operational distinction between yield and proposed cause, together with pre-binding, null-valid inquiry, artifact discrimination, and explicit revision boundaries.

Final statement: incomplete knowledge must remain typed, bounded, trajectory-aware, provenance-preserving, safety-gated, corrigible, and incapable of silently becoming authority.

Version 1.2 is operationally closed, not epistemically final. A contradiction, untyped authority path, unhandled transition, or new consequential distinction automatically reopens the section it defeats.